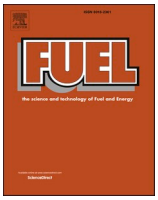
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Biolubricant production from castor oil using iron oxide nanoparticles as an additive: Experimental, modelling and tribological assessment

Uzair Ahmad [a], Salman Raza Naqvi [a,b,*], Imtiaz Ali [c], Faisal Saleem [d], Muhammad Taqi Mehran [a,e], Umair Sikandar [a], Dagmar Juchelková [f]

[a] School of Chemical & Materials Engineering, National University of Sciences & Technology, H-12, Islamabad, Pakistan
[b] National Science and Technology Park, National University of Sciences & Technology, H-12, Islamabad, Pakistan
[c] Department of Chemical and Materials Engineering, King Abdulaziz University, Rabigh
[d] Department of Chemical and Polymer Engineering, University of Engineering and Technology Lahore, Faisalabad Campus, Pakistan
[e] Center for Membrane Separations, Adsorption, Catalysis, and Spectroscopy for Sustainable Solutions (cMACS), KU Leuven (Arenberg), Kapeldreef 75 – box, 2454 3001 Leuven, Belgium
[f] Department of Electronics, Faculty of Electrical Engineering and Computer Science, VŠB – Technical University of Ostrava, 17. listopadu 15/2172, 708 00, Ostrava, Poruba, Czech Republic

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ABSTRACT

The study presents the production of biolubricant from castor oil using Fe_3O_4 nanoparticles and ethylene glycol in a transesterification process, as an additive. Operational parameters such as FAME/alcohol, catalyst loading, and temperature were optimized. The reaction was complete after two hours at 160 °C, giving a yield of 94 %. To enhance the physiochemical properties of modified castor seed oil (MCSO), Fe_3O_4 nanoparticles and ethylene glycol were used. The biolubricant yield was also predicted using artificial neural networks (ANN). The multi-layer perceptron (MLP)-based ANN showed a linear correlation between the output and target values at different temperatures, the amount of catalyst, and the alcohol/FAME ratios during training, testing, and validation. The tribological properties of the biolubricant produced (MCSO + ethylene glycol + 0.5 % Fe_3O_4 nanoparticles) showed the lowest coefficient of friction (almost 50%) and 40% decreased wear compared to raw castor seed oil and other biolubricant samples produced.

1. Introduction

Global energy demand is increasing; despite the harmful impact of fossil fuels on the environment, fossil fuels still constitute the main source of energy [1–3]. Environmental organizations have recently introduced legislation to reduce excessive pollution in the environment [4–6]. These laws are strict in places with significant levels of water and soil pollution [7]. As a result, researchers are focused on developing alternatives to fossil fuel-based products despite the difficulties of high manufacturing costs and performance constraints [8–10].

Vegetable oil-based products are the most significant environmentally friendly products [11,12]. This is due to the remarkable characteristics and characteristics of vegetable oils, including polarity, which implies high lubricity, exceptionally low volatility, high flash point, high viscosity index, and are renewable [13]. The most critical characteristic is that it is biodegradable. Vegetable oil-based lubricants are expected to eventually replace around 90 % of the lubricants used worldwide [14,15].

Numerous oil-bearing plants, including sal, karanja, polanga, neem, mahua, rubber, linseed, and castor, could be investigated for lubrication purposes. However, there is a general lack of research on the extraction of oils from seeds. These seeds can be used as a source of biolubricants. The advantage of these crops is that they can also be grown in dry and semi-arid regions. These fuel sources can be critical components of local economies [16].

Ricinus communis L., a spurge species native to many temperate climates, grows throughout the world. Castor beans contain about 45–60 % oil, and a significant portion of this non-edible oil (80–90 %) is made up of hydroxylated fatty acids (ricinoleic acid), which exhibit unique physicochemical properties, including high specific gravity, viscosity and hydroxyl value [17]. The miscibility of castor oil with ethanol/methanol has given it a unique place in the esterification process using enzymes. The inhibitory effect of the alcoholic substrate on lipase is not

* Corresponding author.
E-mail address: salman.raza@scme.nust.edu.pk (S. Raza Naqvi).

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Nomenclature

ANN	Artificial neural network
ANS	Automated network search
BFGS	Broyden-Fletcher-Goldfarb-Shanno
CSO	Castor seed oil
FAME	Fatty acid methyl ester
FTIR	Fourier transform infrared spectroscopy
MCSO	Modified castor seed oil
MLP	Multilayer perceptron
MUFA	Monounsaturated fatty acid
RCSO	Raw castor seed oil
SANN	Statistica automatic neural networks

significant, and this behavior facilitates the advancement of the reaction [18]. As a result, castor oil was identified as a preferred raw source for the production of biolubricants [19]. The scientific name of the castor seed oil is Ricinus communis, which somehow belongs to the Euphorbiaceae family. South Asia and East Africa are its natives and it has spread to many topological regions of the world. The shape of castor is like beans, lacks volatility, and has the light-yellow oil. Due to the considerable amount of monounsaturated fatty acid (MUFA) content, it has larger thermal oxidative stability, and it is widely used in industries with high-temperature conditions as well. On average, castor contains about 45–60% oil content. In 30 different countries, the castor plant can grow, including Brazil, India, Russia, China, the United States, and Thailand, and these are the countries that can make up to 85% of the castor oil production.

The production of castor oil on a global scale is estimated to be 1.8 Mt/year in the following countries: Brazil, India, and China [18]. However, the United States of America, the European Union, and Japan are the main importers of castor oil [20]. This is an inedible oil that has an important value in the chemical and physical fields. As a result of the presence of ricinolein acid, castor oil has many physical and chemical properties. Castor oil contains hydroxy, carboxylate, long chain, and unsaturation hydrocarbons present in ricinoleic acid that make it unique. As a result of its unique properties, this oil has great potential as a petroleum-based product and is good for biorefinery. There are many applications of castor oil, such as wax, pharmaceutical industries, cosmetics, care products, paints and coatings, plasticizers, and biolubricants [21].

Chemical modifications such as transesterification, epoxidation, etc. are frequently used to improve the suitability of the lubricant for lubricating needs. Chemical modification can be used to improve the lubricating characteristics of vegetable-based lubricants [22]. Solid powder nanoparticles are more effective additions to vegetable oils because they increase their lubricity and make them suitable for tribological applications [23,24]. Iron oxide (Fe_3O_4), a solid additive with a crystalline structure, is one of the solid additives [25]. Numerous studies have been conducted on metallic oxide nanoparticles as lubricant additives, including those on TiO_2 [26,27], SiO_2 [28,29], CuO [30] and ZnO [23,31]. Furthermore, a low nanoparticle concentration is preferable to improve the lubrication of rubbing surfaces (preferably below 1.5 wt%). According to the authors' information, the utilization of iron oxide nanoparticles as an additive in the transesterification process for the production of improved biolubricants has not been thoroughly investigated before. Therefore, the authors aim to address this research question in this study.

Artificial neural networks (ANNs) are robust computational means for solving problems, particularly in biotechnology, where the underlying theory is less obvious but the data is accessible [32,33]. ANN classification may involve one or more of the following ANN characteristics: the direction of information flow inside the network, the connection and extent of the network's neurons, the type of learning algorithm, and the learning rule [34,35]. ANNs are capable of training complex datasets [36,37]. In addition, this work presents the application of multilayer perceptron (MLP)-based ANN to predict the biolubricant yield as a function of input process parameters such as temperature, catalyst loading, and alcohol/FAME ratio for different time durations. The tribological properties of the biolubricant samples produced were also analyzed. It is the study of surfaces that interact in relative motion and includes the science of lubricant, friction, and wear. Since the start of the 20th century, knowledge in all aspects of tribology has exploded, owing to immense industrial growth, which results in the demand for a better understanding of tribology.

2. Materials and methods

2.1. Materials:

Methyl alcohol was purchased from a local vendor. Iron oxide (99 % purity) nanoparticles and ethylene glycol were obtained from the vendor as an additive. Castor seed oil (CSO) was obtained from the local market; and isooctyl alcohol was obtained from a local chemical supplier.

2.2. Analysis of raw castor oil

The raw castor seed oil was obtained from the vendor Index. The chemical modification process of castor oil to produce a biolubricant was carried out in three stages. These stages were the calculation of free fatty acids (FFAs), the production of fatty acid methyl esters (FAME), and the third and last step was the production of biolubricant from the produced FAME. The first step was to calculate the FFA concentration of castor seed oil (CSO), which should be less than 1 %. For this, the CSO was immersed in sodium hydroxide (NaOH) and methanol (CH_3OH) and then heated to 70 °C. Subsequently, 1 g of warmed oil, 50 ml of propyl alcohol, and six phenolphthalein spatters were squeezed and mixed in a test tube. After completion, a droplet of NaOH was mixed into a jar until the solution became pink. The proportion of FFA represented in CSO was calculated using eq. 1.

$$\%FFA = \frac{\text{volume of alkali(ml)} \times \text{Alkali concentration(N)} \times 25.6}{\text{mass of oil sample (g)}} \quad (1)$$

To find the functional groups of the raw castor oil, Fourier transform infrared spectroscopy (FTIR) was performed. Fig. 1 shows the FTIR spectrum of castor seed oil. The castor oil shows a typical ester carbonyl

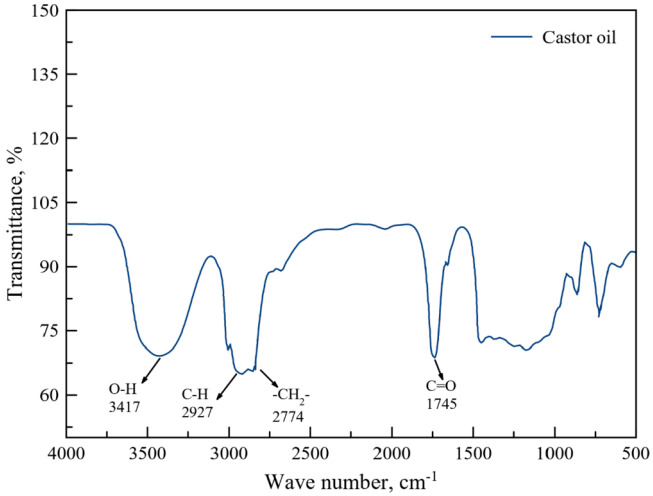


Fig. 1. FTIR spectrum of castor seed oil.

(C = O) stretching of triglyceride at 1745 cm^{-1}, and at 2774/2927 cm^{-1} is due to the symmetrical and asymmetrical vibrations of aliphatic ($–CH_2$) fatty acid chain. Castor oil also shows a hydroxyl (O–H) band stretching between 3210 and 3640 cm^{-1} arising from the hydroxylated ricinolic acid. Broader peak with large area suggests the presence of hydroxyl group containing constituent with relative high concentration.

2.3. Development of a biobased lubricant

The fatty acid methyl ester (FAME) was produced via a transesterification procedure using a base catalyst. Fig. 2 presents the experimental setup of the transesterification process. The transesterification procedure was carried out in a double-neck round bottom flask and sealed with a Graham condenser. Oil and alcohol; methanol with a molar ratio of 1:6 was pumped into a collar flask and sunk into a dehydrator. A 1-percent potassium hydroxide (KOH) substrate was introduced into the oil to improve the interaction between the mixes. The reaction temperature was kept stable at 60 °C for 1 h in the dehydrator. After that, the sample was poured into a separate funnel and allowed to settle to differentiate between FAME and glycerol. The bulky material, which was glycerol, would sink to the base, while the FAME remained at the top. Therefore, glucose was extracted, and FAME was passed through a cleaning procedure. This method was carried out by pumping warm water into the FAME in the measuring cylinder. Therefore, such mixes are stirred with the introduction of phosphoric acid until the mixture reaches pH 7. The samples were allowed to remain at a temperature in the range of 120 °C to eliminate additional methanol and water material.

Finally, to develop a biolubricant, the FAME developed in the first step of transesterification was used. The process was carried out in a 1000 ml double-neck round bottom flask with a thermometer, a measuring outlet, and a Graham condenser. The flask was placed on a heating mantle at a temperature of 160 °C. FAME and isooctyl alcohol were pumped into the round bottom double neck flask and steadily heated to 160 °C. Electromagnetic mixers have been utilized to ensure that both formulations have been adequately blended. Sodium methoxide was used as a precursor and mixed with a solvent for 2 h to finish the process and become a biolubricant known as modified castor seed oil (MCSO). After the transesterification process, the MCSO was mixed with additives. The additives used were ethylene glycol and iron oxide (Fe_3O_4) nanoparticles. The surface modulation device was used to ensure the stabilization of the additives with the modified liquid. For a consistent suspension of the additives, they are mixed with a magnetic stirrer of 750 rpm for a duration of 3.5 h. The mixture was stirred again through the ultrasonic sonicator for 1 h. Table 1 illustrates the samples prepared with and without additives. The complete schematic procedure is illustrated in Fig. 3 which explains the experimental work performed in this entire paper. Raw castor seed oil is first transesterified in the presence of a KOH catalyst (1% w/v) at 60 °C and the oil - alcohol ratio was 1: 6. After the transesterification process, the fatty acid methyl ester (FAME) was separated from the glycerol. The produced FAME was again transesterified at 120 °C and the FAME to alcohol ratio was 1:2. At this stage, 0.5% Fe_3O_4 nanoparticles were also added to improve the properties of a biolubricant. Once the biolubricant was produced, physiochemical and tribological analyses were done to find its properties.

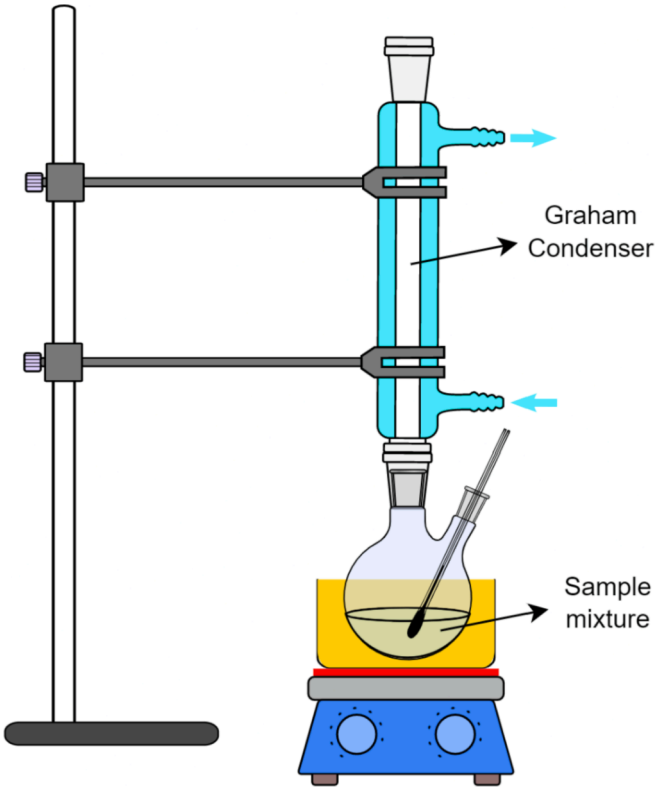


Fig. 2. Experimental setup of the transesterification process.

2.4. Methodology for artificial neural networks

Artificial neural networks (ANN) are useful tools for nonlinear regression. An advantage of ANNs is that they can be trained on any data set without prior knowledge of the modelled phenomenon. In this study, ANN based on multilayer perceptron (MLP) is used for non-linear regression. MLP consists of input, hidden layer, and output layers. The hidden and output layers are made up of artificial neurons. The input layer transfers the input signals through the hidden layers to the output layer in feedforward mode. Hidden and output layers also include a bias term. The connection weights were then adjusted using an error back-propagation algorithm while training from the randomly assigned initial values. During training, the error between the target and output signals is minimized by adjusting the number of neurons in the hidden layers and the transfer functions of the neurons. For this study, the Statistica Automatic Neural Networks (SANN) present in the TIBCO Statistica 13.5 software was used. SANN uses the Broyden-Fletcher-Goldfarb-Shanno (BFGS) iterative algorithm to minimize the error function of ANN [38]. Commonly, sum of squares error function (SSE) is used as an error function which by definition computes sum of squared difference between the output and target values and is scaled by 1/2N.

$$SSE = \frac{1}{2N}\sum_{i=1}^{N}(y_i - t_i)^2 \quad (2)$$

where N is the no. of training instances, t_i is the target values and y_i is the predicted output of the ith instance in the training data set.

The Automated Network Search (ANS) feature of SANN compares various networks and selects five models based on the best performance criteria, saving time by avoiding the lengthy trial-and-error process. The activation functions used in ANS for the hidden and output layers are identity, tanh, logistic sigmoid, exponential, Softmax, sine, and Gaussian are given as eqs. (3–9).

Identity:	a	(3)
Hyperbolic tangent:	$\frac{e^a + e^{-a}}{e^a + e^{-a}}$	(4)
Logistic sigmoid:	$\frac{1}{1 + e^{-a}}$	(5)
Exponential:	e^{-a}	(6)
Softmax:	$\frac{e^{a_i}}{\sum e^{a_i}}$	(7)
Sine:	$\sin(a)$	(8)
Gaussian:	$\frac{1}{\sqrt{2\pi\sigma}} e^{\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]}$	(9)

Table 1
Samples prepared for analysis.

Symbol	Description	Sample
RCSO	Raw Castor Seed Oil	
MCSO	Modified Castor Seed Oil	
MCSO + 0.5 % Fe_3O_4 nanoparticles	Modified Castor Seed Oil with 0.5 % iron oxide nanoparticles	
MCSO + ethylene glycol	Modified Castor seed oil with ethylene glycol	
MCSO + ethylene glycol + 0.5 % Fe_3O_4 nanoparticles	Modified castor seed oil with ethylene glycol and 0.5 % iron oxide nanoparticles.	

where *a* is the weighted sum of neuron's input.

The flow chart for the ANN modelling is shown in Fig. 4. The relative influence of the input variables on the output parameter is assessed in terms of relative sensitivity. ANN based on MLP has been used to model, optimize, and predict the yield of biofuels [39–41].

3. Results and discussion

3.1. Effect of temperature on yield

As shown in Fig. 5, the biolubricant yield was the lowest at 150 °C, giving only 83 % at the end of the experiment. For 160 °C and 170 °C, the yields increased to 90 % at 100 min. At the end of the experiment, the yield was 94.5 % and 94.8 % at 160 °C and 170 °C, respectively. As the temperature increases, the reaction conversion also increases; hence, the biolubricant yield also increases. Thus, a distinct effect of temperature was observed on the final yield and kinetics at three different temperatures. These trends were very similar to those of previous studies. [42,43].

3.2. Effect of catalyst concentration on yield

The effect of the catalyst concentration was determined at 160 °C, under the assumption that the catalyst concentration would be more effective at lower temperatures (Fig. 6). From a kinetic point of view, an increase in the catalyst concentration increased the reaction rate. Taking into account the biolubricant yield, the final conversion was too low (less than 80 %) using a 0.5 % catalyst (w/v) compared to 1 % and 2 %. (94.5 and 95 %, respectively). For 1 and 2 % of the catalyst loading, the completion was achieved only after 2 h, as shown by an asymptote. There seems to be a catalyst concentration limit beyond which no increase in conversion was detected. As a result, a concentration of 1 % w/v of the catalyst was observed to be the most viable to obtain the maximum yield. Other researchers have reported similar findings [44,45].

3.3. Effect of alcohol/FAME on yield

Regarding the molar ratio of Iso octyl alcohol/FAME, there were differences, especially for the equimolar concentration (1:1) (Fig. 7). With this value, a yield of hardly 80.5 % was obtained, while for molar ratios of 2:1 and 3:1, the yield improved significantly (94.5 % and 95 %, respectively). These findings revealed that the alcohol content had a significant effect on the equilibrium, rerouting the process toward the production of the biolubricant. In terms of kinetics, the alcohol content had a favorable effect, increasing the rate of creation of the product. As a result, some authors recommend using an amount of alcohol greater than a 1:1 M ratio [46,47]. In this study, the most suitable molar ratio was found to be 2:1.

In summary, considering the relationship between economic factors and yield, the following values were chosen for biolubricant production: 160 °C temperature, 1 % catalyst (w/v) and a 2:1 M ratio of alcohol/FAME.

3.4. Performance prediction using artificial neural networks

The output parameter (bio lubricant yield %) was modelled for the input (temperature, catalyst loading, and alcohol/FAME ratio) [48]. To design the neural network, the data set was randomly distributed into three sets of training (70 %), testing (15 %), and validation (15 %). Table 2 lists the best neural model structure with quality metrics during training, testing, and validation. The model structure represents the corresponding neurons used in the input, hidden, and output layers, respectively. The nodes in the input layer are equal to the input variables of the model, which are two in this case. The number of neurons and the activation function in the hidden layer were obtained by the BFGS iterative algorithm for better predictability. The only neuron in the output corresponds to the target.

Fig. 8 shows the correlation of the output with the target values. Models showed good data fit during ANN training, testing, and validation. ANN architectures were trained to predict biolubricant yield. The five-best MLP-based ANN structure to predict the effect of temperature

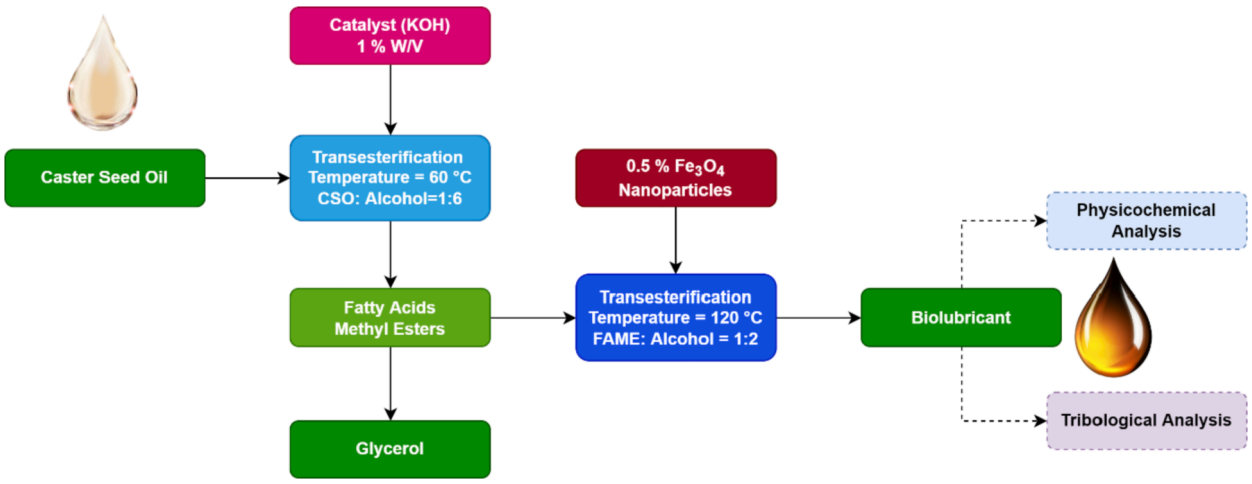


Fig. 3. Experimental setup of research methodology.

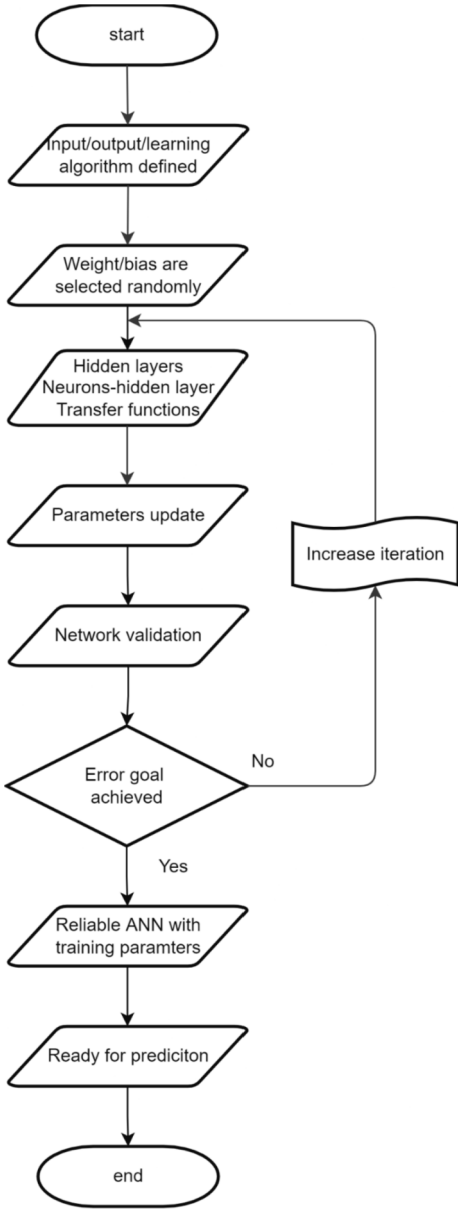


Fig. 4. Generic flow chart for ANN modeling.

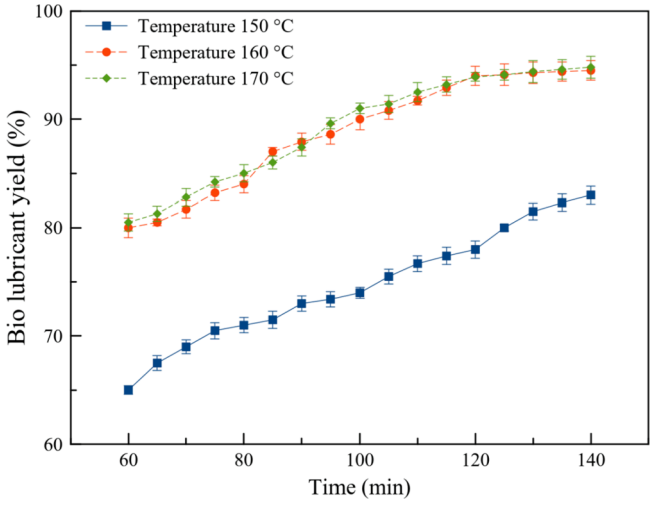


Fig. 5. Biolubricant production at different temperatures (Catalyst 1 % w/v and alcohol to FAME ratio of 2:1).

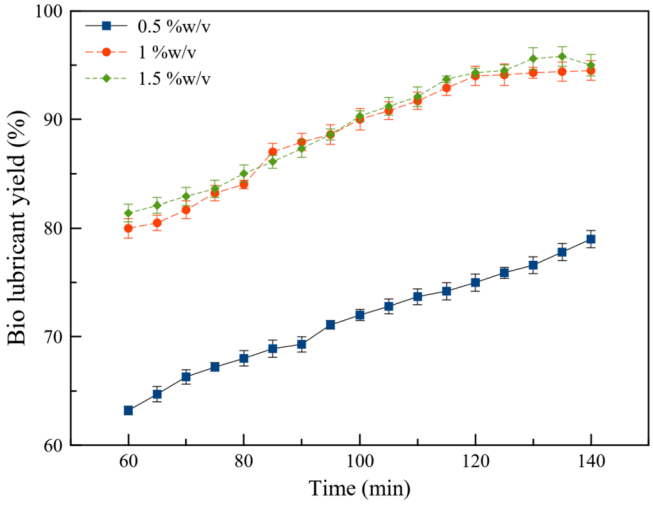


Fig. 6. Biolubricant production as a function of catalyst concentration (Temp 160 °C and alcohol to FAME 2:1).

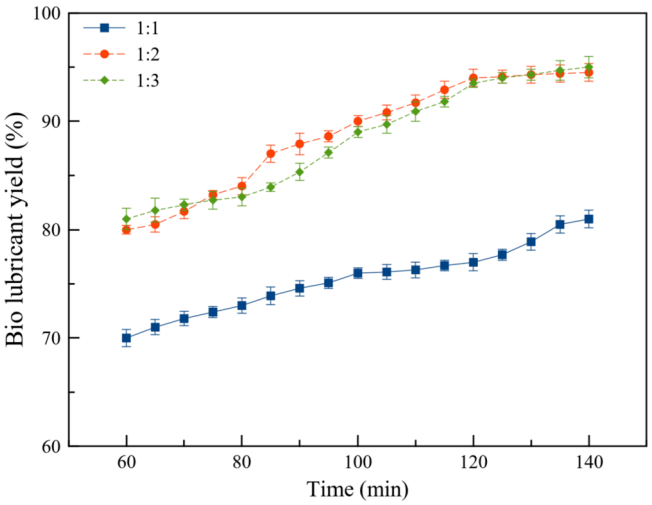


Fig. 7. Biolubricant production according to the alcohol/FAME molar ratio (Temp 160 °C and catalyst 1 % w/v).

on biolubricant yield required 3–11 neurons in the hidden layer with tanh and exponential as activation functions. A higher R value and low error during training, testing, and validation of the data set make these structures best suited for prediction. MLP-based ANN with 5–11 hidden layers using tanh, exponential, and logistic functions performed well in predicting the effect of catalyst loading.

The biolubricant yield as a function of the alcohol/FAME ratio was best described by a model containing 3–10 hidden layers using tanh and logistic as the activation function. The best models were used for the sensitivity analysis. The results are presented as a relative influence of the input parameters on the yield of the biolubricant. As shown in Fig. 9, catalyst loading and the alcohol/FAME ratio influence the yield equally by 38 % compared to temperature, which influences by 24 %.

Generated model codes were used to verify the accuracy of the internally trained, tested, and validated models using the complete data set. Fig. 10 (a) shows the comparison of the experimental and simulated biolubricant yield values at different times and at different temperatures. It is evident that the model accurately simulated the measured values. Similar observations can be made for the influence of catalyst loading and alcohol/FAME ratio in Fig. 10 **(b&c).**

4. Biolubricant characterization

The characteristics of each lubricant sample were determined and confirmed according to the American Society of Testing and Materials (ASTM). To evaluate the buoyancy of the oils, density tests were performed. The samples were weighed using a pycnometer at a temperature of 20 °C according to ASTM 4502. The viscosity was determined using an Ostwald viscometer and according to ASTM D445 [49] at 40 °C and 100 °C for the oil. The flashpoint was defined as the temperature at which a sample of oil began to vaporize and form an ignitable mixture in the air. It was determined using an ASTM D92 [50]

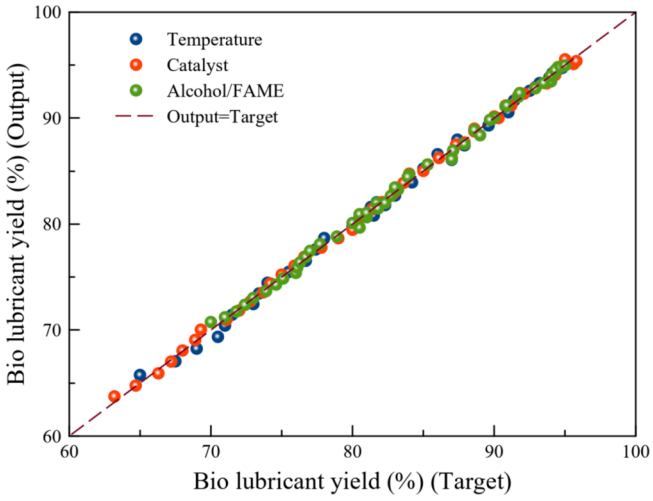


Fig. 8. Correlation between output and target for different input parameters with ensemble best-fit MLP-based ANN.

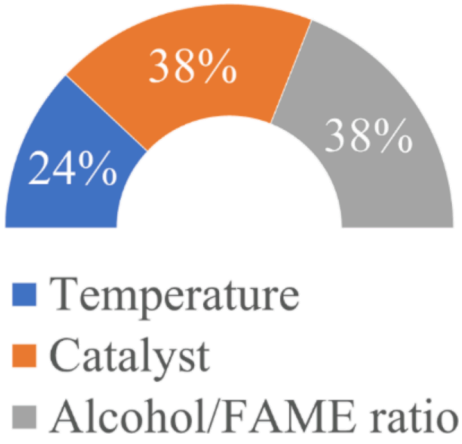


Fig. 9. Relative sensitivity of input variables to biolubricant oil yield.

Table 2
Structure and performance metrics of the five best neural networks for each variable.

Variables	Structure	Train		Test		Validation		Activation Functions	
		R	Error	R	Error	R	Error	Hidden Layer	Output Layer
Temperature	2–11-1	0.999183	0.057807	0.998997	0.057807	0.999848	0.043174	Tanh	Tanh
	2–11-1	0.999086	0.064494	0.998581	0.071514	0.999439	0.094270	Logistic	Tanh
	2–7-1	0.998518	0.104536	0.998689	0.068659	0.998826	0.189274	Exponential	Tanh
	2–3-1	0.998720	0.090927	0.998250	0.076732	0.999286	0.169229	Tanh	Tanh
	2–3-1	0.997338	0.187546	0.996087	0.145784	0.998399	0.423223	Tanh	Identity
Catalyst	2–11-1	0.999444	0.050274	0.999442	0.048043	0.999483	0.060261	Tanh	Logistic
	2–5-1	0.998610	0.140608	0.999294	0.081079	0.999460	0.065348	Tanh	Logistic
	2–11-1	0.999241	0.069111	0.999756	0.017111	0.999277	0.101026	Exponential	Tanh
	2–9-1	0.998830	0.105978	0.999873	0.015749	0.999025	0.096932	Logistic	Identity
	2–11-1	0.999565	0.039480	0.999830	0.024003	0.999290	0.079727	Tanh	Exponential
Alcohol /FAME ratio	2–4-1	0.999021	0.059502	0.997975	0.083490	0.998777	0.074705	Logistic	Logistic
	2–3-1	0.997971	0.120359	0.997364	0.090246	0.997868	0.140045	Tanh	Sine
	2–10-1	0.998627	0.081591	0.999332	0.124295	0.998301	0.098053	Tanh	Logistic
	2–3-1	0.998107	0.111453	0.999854	0.021593	0.999114	0.081839	Tanh	Identity
	2–6-1	0.998767	0.073367	0.999772	0.057083	0.997935	0.114007	Tanh	Logistic

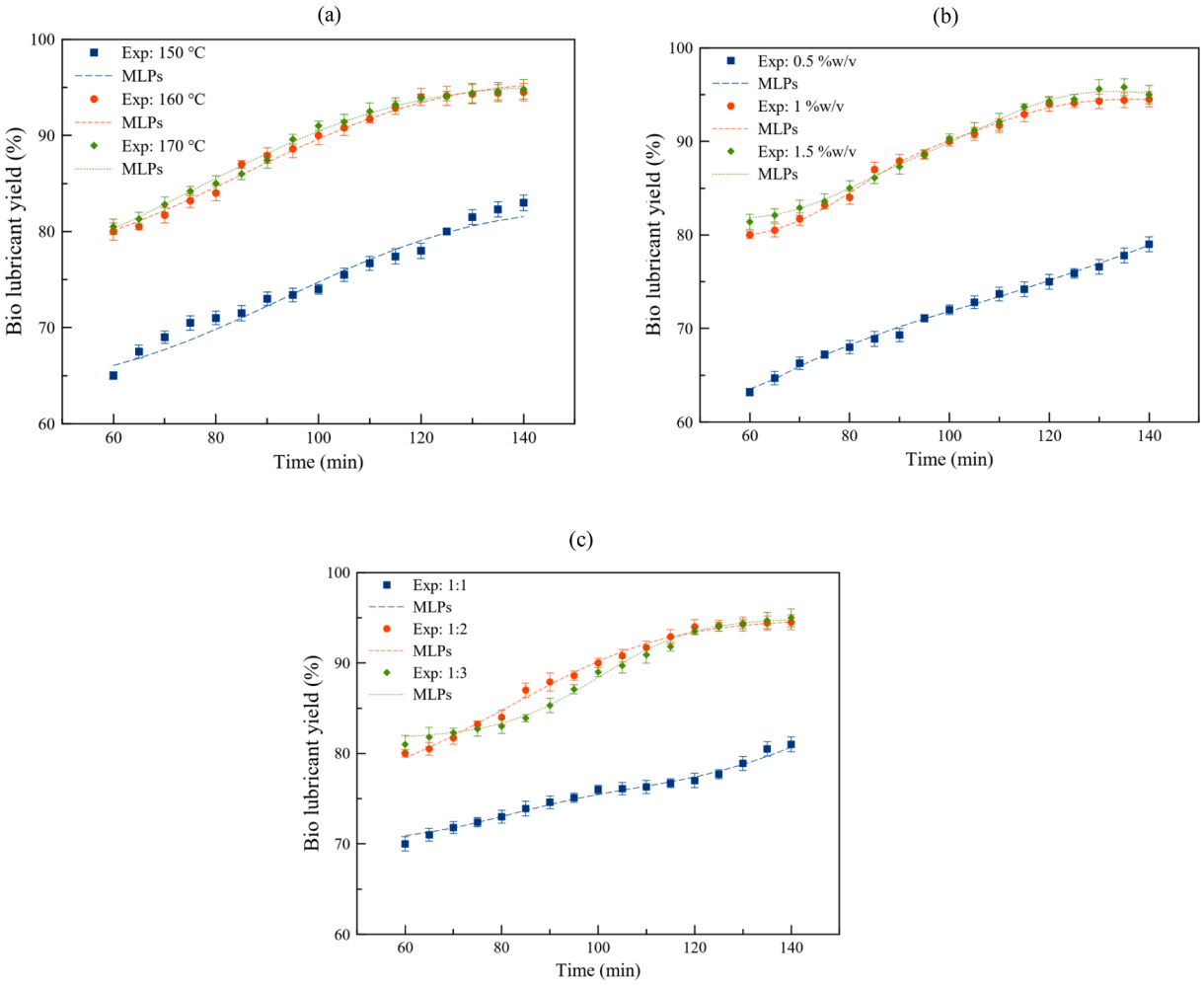


Fig. 10. Biolubricant yields measured and simulated at different (a) temperatures, (b) catalyst loading, and (c) alcohol/FAME ratios.

closed cup flash point tester (Pensky-Martens). Lubricant samples were placed in the cup, heated, and then regularly opened to initiate ignition. According to the ATSM D-97 standard [51], the pour point of the samples was measured. The synthesized biolubricant was analyzed in terms of density, flash point, pour point, and kinematic viscosity at 40 °C and 100 °C. The results of the lubricants were compared with those of the commercial lubricants (75 W-140 and AG100). We can see a decrease in the viscosity values as the temperature is increased. Table 3 shows the physicochemical properties of all lubricants. It is quite clear that MCSO with ethylene glycol and 0.5 % Fe_3O_4 nanoparticles shows a better flash point result compared to MCSO with 0.5 % Fe_3O_4 nanoparticles.

Fig. 11 illustrates the relationship between kinematic viscosity and temperature for all lubricant samples. Most lubricants showed similar patterns. Under such circumstances, the viscosity decreased as the temperature increased. Repetitive contact with rubbing surfaces reduced the useful life of lubricating chemicals. It may lead to the formation of carbonaceous deposits, induce wear, and impair the performance of the device. It must be recognised that higher temperatures are required at the molecular level to break covalent bonds, initiate free-radical processes, and cause the viscosity of the lubricating material to decline. The kinematic viscosity of (75 W-140 and AG100) was found to be 216 cSt and 175 cSt at 40 °C [53]. However, RCSO and MSCO show better kinematic viscosity at 40 °C (193 and 210 cSt, respectively) than the commercial lubricant AG100. Fig. 11 shows that with the addition of 0.5 % Fe_3O_4 nanoparticles to MCSO, the kinematic viscosity increases compared to RCSO and MCSO. The addition of ethylene glycol and 0.5 % Fe_3O_4 nanoparticles to MCSO shows the highest value of kinematic viscosity.

Table 3
Physicochemical properties of lubricant samples.

Oil Type	Density (kg/m^3) at 20 (°C)	Flash Point (°C)	Pour point (°C)	Kinematic viscosity cSt at 40 (°C)	Kinematic viscosity cSt at 100 (°C)	Ref.
Raw Castor Oil	959 ± 8	147 ± 0.5	−5 ± 0.2	193 ± 4.8	20 ± 0.9	This
Modified Lubricant	890 ± 5	243 ± 1	−15 ± 0.4	210 ± 6.1	22 ± 1.2	study
Modified Lubricant + Fe_3O_4	894 ± 7	266 ± 2	−16 ± 0.5	216 ± 7.2	25 ± 1.5	
Modified Lubricant + Ethylene Glycol (EG)	891 ± 6	259 ± 1	−14 ± 0.3	213 ± 6.2	23 ± 1.2	
Modified Lubricant + Fe_3O_4 + EG	893 ± 6	272 ± 2	−18 ± 0.6	218 ± 7.1	27 ± 1.8	
AG100		244	−18	216	19.6	[52]
75 W-140		174	−54	175	24.7	[53]

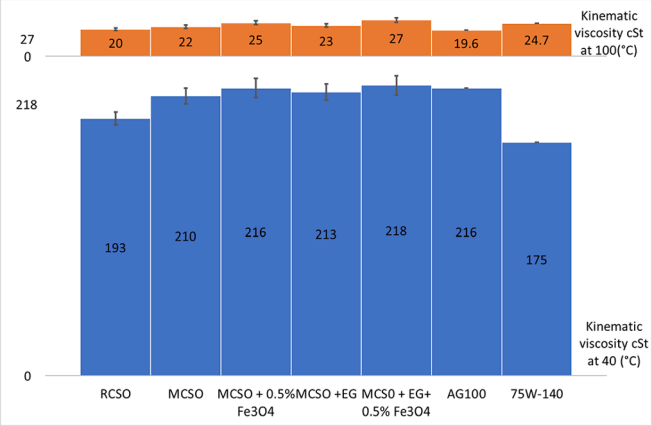


Fig. 11. Kinematic viscosity of lubricant samples.

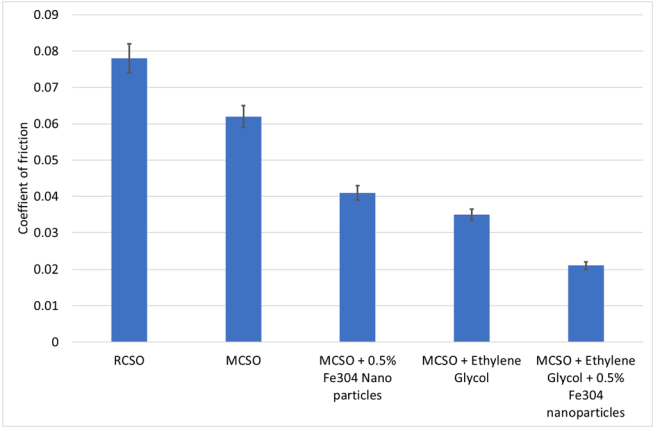


Fig. 13. Co-efficient of friction of different biolubricant samples.

4.1. Tribological analysis

The pin-on-disc machine has been used to explore wear and friction parameters during lubricant deployment. The friction force was measured in the operating application of the device using the load cell coupled to the tribo machine. To obtain the friction coefficient, the friction forces were normalized by the applied load. The lubricant was supplied to the interface dropwise with the help of a pump powered by an electric motor (Fig. 12).

4.2. Coefficient of friction

The coefficient of friction for various lubricant samples is shown in Fig. 13. Esters have been shown to develop in the formation of an effective layer on the surface during movement. The presence of long-chain molecules strengthens the load capacity of the lubricating coating [54]. Improvement in the results has been observed in the case of additive addition. With the addition of additives, ethylene glycol and Fe_3O_4 nanoparticles in MCSO showed better friction coefficient results compared to RCSO and MCSO. Chemically treated vegetable oil provides better lubrication and enhanced film adsorption on metallic surfaces. The adsorption coating reduced friction between the surfaces, resulting in a decrease in cutting forces.

4.3. Wear

To determine pin wear, the samples were thoroughly cleaned with acetone and dried in a preheated oven prior to weighing. Fig. 14 clearly

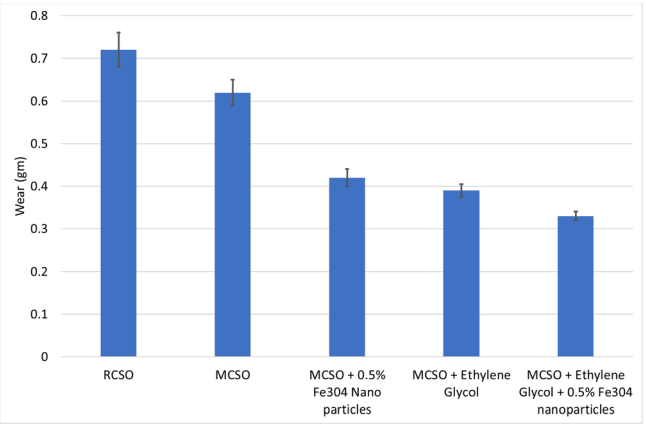


Fig. 14. Wear material for biolubricant samples produced.

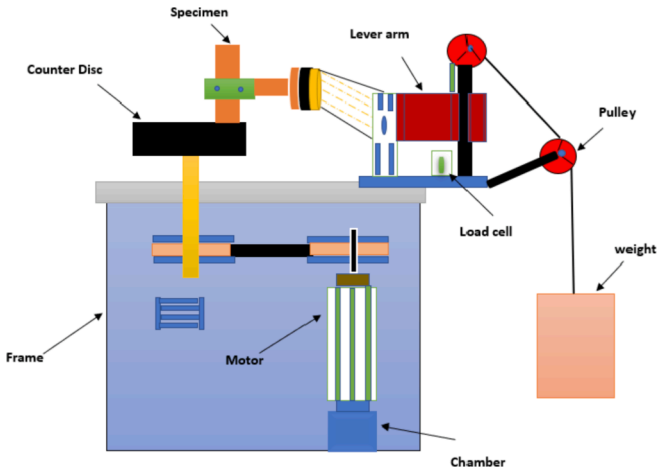


Fig. 12. Experimental setup of tribological analysis.

illustrates the wear of the pin. RCSO and MCSO show almost the same wear behavior. The additives ethylene glycol and Fe_3O_4 nanoparticles show better wear compared to RCSO and MCSO. The effect of nanoparticles on the corrosion and wear characteristics depends on the dispersion of the particles within the epoxydated oil. Fe_3O_4 nanoparticles have a greater ability to disperse in solution, allowing the anti-wear mechanism to function properly. The tricarbamyl alcohol reaction occurs between the transcription factors of the components and the metal atoms during their interaction, resulting in the formation of an extraordinarily excellent lubricant coating. The better performance of MCSO + ethylene glycol + 0.5 % Fe_3O_4 nanoparticles is mainly due to the production of thin lubricating films with metals in contact. The creation of a thin molecular lubricant coating helped to reduce friction, thus preventing premature material wear. The development of a protective lubricating layer on the surfaces was anticipated to reduce the forces operating on them.

5. Conclusions

Biolubricant preparation (conversion and kinetics) was affected by the FAME/alcohol molar ratio, catalyst concentration (Fe_3O_4) and temperature. Thus, the temperature increased the final yield with kinetic changes at higher temperatures. As the molar ratio of FAME/alcohol increased, the yield improved. Similarly, as the catalyst loading increased from low to intermediate values, the ultimate conversion increased. To optimize the yield and economics, intermediate temperature and catalyst concentrations (T = 160 °C, catalyst = 1 % w/v) were used, as well as the maximum alcohol/FAME ratio (2:1). Once the critical conditions for biolubricant production were established, the

yield of the castor oils obtained was acceptable, approximately 94 %. When additives are added to the MCSO, the physiochemical properties of the biolubricants produced are enhanced. MCSO + ethylene glycol + 0.5 % Fe_3O_4 nanoparticles show the best results among the other biolubricant samples produced. Training, testing and validation of the artificial neural network (ANN) showed a high linear regression coefficient (more than 0.997) for the biolubricant yield as a function of temperature, catalyst loading, and alcohol/FAME ratio. The results obtained from the developed ANN models were found to agree well and, hence, it can be concluded that the ANN models can effectively predict the biolubricant yield for varying temperatures, catalyst loading, and alcohol/FAME ratios for different time durations. The relative sensitivity of the temperature to the biolubricant yield was found to be lower than that of the catalyst loading and alcohol/FAME ratios.

CRediT authorship contribution statement

Uzair Ahmad: Conceptualization, Writing – original draft. **Salman Raza Naqvi:** Methodology, Writing – original draft, Validation, Supervision, Resources, Project administration. **Imtiaz Ali:** Writing – original draft, Formal analysis. **Faisal Saleem:** Writing – review & editing, Visualization. **Muhammad Taqi Mehran:** Writing – original draft, Data curation, Validation. **Umair Sikandar:** Investigation, Writing – review & editing. **Dagmar Juchelková:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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